Politecnico di Torino

Qubit Electronics
Lab 4

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1 Introduction

During this laboratory session, we will focus on meshing and resolution concepts using the Poisson equation. The experimental subject under investigation is a real double quantum dot device, where confinement is achieved by exploiting Silicon-Silicon Dioxide (Si-SiO2) materials. Download the `Lab4.zip` from the Portale della Didattica.

2 Mesh

Open the file `DQD-SiMOS.geo` and report the dimensions of the structure (x, y, z).

Execute the mesh command

```
gmsh DQD-SiMOS.geo
```

following the procedure from the prior lab session. Click on `Tools->Message console` and do not close the window until the compilation is over.
In case you are using the server, the mesh is not shown and it is necessary to use the command

```
gmsh -3 DQD-SiMOS.geo
```

to compile it. Otherwise, if you are using your PC, it is necessary to download GMSH.

Considering that the time required to compile the mesh can be very long, it is suggested to compile it at home.

Question: Determine the regions exhibiting greater density. Analyze various sets of regions grouped by "region_{name_region}". Focus on regions labeled with names starting with "region_quantum".

Question: Identify specific indicators of this varying density within the code, corresponding to the graph recently examined.

3 Poisson

Open the file `SiMOS_poisson.py` and meticulously review the code.

Question: Determine the applied voltages across distinct regions and comprehend the operational temperature.

Question: Explain the different materials listed in the code.

Review the following code:

```
params_poisson = PoissonSolverParams()
params_poisson.tol = 1e-7
params_poisson.initial_ref_factor = 0.55
params_poisson.final_ref_factor = 0.85
params_poisson.min_nodes = 0
```

```
params_poisson.maxiter = 1000
params_poisson.dot_region = dot_region
params_poisson.h_dot = 0.3
params_poisson.max_nodes = 1e10
```

Question: Report what the parameters set in the previous code represent.

In case you are working on the LED PC remember to copy the documents inside `qtcad.zip`, otherwise it is not necessary.

Then, launch the command:

```
python SiMOS_poisson.py
```

4 Paraview

Open Paraview and import the file ending with EC.vtu from the "Output" folder created after launching `python SiMOS_poisson.py`.

File EC.vtu: Open Paraview, import the file, click "Apply," then "Slice" with Z-normal at a height of 45. Finally, click "Coloring" and "Rescale to data range".

Question: Explain the areas present in the cut and the associated colors.

Click on "Plot Overline," and vary the variable x across the domain, with y values set to 136.5 and z remaining at 45.

Question: Describe the graph, showing how the potential changes in the domain and what the minima represent. Explain if this behavior in the central region is as expected.